CAMON: Low-Cost Silicon Photonic Chiplet for Manycore Processors

Zhehui Wang, *Member, IEEE*, Zhifei Wang, *Student Member, IEEE*, Jiang Xu, *Member, IEEE*, Yi-Shing Chang, *Member, IEEE*, Jun Feng, *Student Member, IEEE*, Xuanqi Chen, *Student Member, IEEE*, Shixi Chen, *Student Member, IEEE*, and Jiaxu Zhang, *Student Member, IEEE*

***Abstract*—While many new applications prefer manycore processor with a large number of cores, the exploding communications among multiple cores, caches, and off-chip memories is posing a fundamental challenge on manycore designs. Silicon photonics-based interconnection network promises high bandwidth, low latency, and high energy efficiency, and can potentially meet the communication requirements of manycore processors. In this paper, we propose CAMON, a small low-cost silicon photonic chiplet integrated into the manycore processor package. CAMON chiplet can effectively alleviate the communication bottlenecks of manycore processors and improve the energy efficiency of data movement, especially for large-scale systems. We develop a distributed arbitration system, a low-power low-latency optical interface, and an off-chip laser preactivation mechanism for CAMON. The experimental results show that compared with the electrical network, CAMON can improve the full-system performance per energy by 4.6×, speedup the manycore processor by 2.7×, and save the area of the processor die by 3%, in a 512-core system.**

***Index Terms*—Cache and memory network, manycore processor, silicon photonic chiplet.**

I. Introduction

TODAY, manycore processor integrates a large number of cores. The bandwidth demand among processor cores, caches, and off-chip memories is continuously increasing. The cache and memory network will become a critical component in manycore processors. State-of-the-art manycore processors use electrical network-on-chip (NoC) for on-chip communication, and electrical interconnects for off-chip communication. In future manycore processors, the exploding communications are posing a fundamental challenge on manycore designs. The processor cores have stronger processing capacities, making them read and write data more frequently. As more processor cores are integrated in a single chip and working in parallel, the total memory throughput increase. Besides, today's application has larger data sets to process. The cache and memory network should have higher throughput and higher energy efficiency. However, if the system scale keeps growing, it is difficult to keep the latency and energy consumption of the electrical NoC at a reasonable level. It is also hard to effectively increase the total bandwidth of electrical interchip interconnects because of the limited space for pins on the package substrate [1].

Silicon photonics is one of the promising technologies for future manycore processors. Building a cache and memory network using optical links shows three potential advantages [2]. First, the optical interconnect has ultra higher bandwidth densities than electrical interconnects, especially when the wavelength-division multiplexing (WDM) technology is applied. This feature makes the optical network more suitable for the heavy traffic of manycore processors. Second, the energy consumption of optical interconnects is relatively independent to the traversal distance. Board-level optical interconnect shows much higher energy efficiency than the electrical one. Third, optical pins have larger bandwidth density than electrical pins [3]. Using optical interconnects for interchip communication is an effective way to alleviate the chip pin constraint on the package.

The data rate of optical interconnect with a single wavelength is constrained by the speed of electrical circuits. Using the WDM technology, the data rate of optical interconnect is proportional to the number of optical wavelengths. The exact bandwidth density of optical interconnect is much higher than that of electrical interconnect. There are two parameters evaluating the bandwidth density. The first parameter is called the area bandwidth density. This parameter evaluates the data throughput efficiency through the chip package. The area bandwidth density of optical interconnect is in the order of 10^4 Gb/s/mm^2 [2], while that of electrical interconnect is in the order of 10^2 Gb/s/mm^2. The second parameter is called the linear bandwidth density. This parameter evaluates the data throughput efficiency through parallel interconnects. The linear bandwidth density of optical interconnect is in the order of 10^4 Gb/s/mm, while that of electrical interconnect is in the order of 10^2 Gb/s/mm.

We propose CAMON, a low-cost silicon photonic chiplet integrated into the processor package. It can effectively alleviate the communication bottlenecks of manycore processors, and improve the energy efficiency of data movement, especially for large-scale systems. It is mounted on a top of the

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Z. Wang, Z. Wang, J. Xu, J. Feng, X. Chen, S. Chen, and J. Zhang are with the Department of Electronic and Computer Engineering, Hong Kong University of Science of Technology, Hong Kong (e-mail: zhehui@connect.ust.hk; jiang.xu@ust.hk).

Y.-S. Chang is with the Silicon Photonics Product Division, CG/DCG, Intel Corporation, Santa Clara, CA 95054 USA (e-mail: yi-shing.chang@intel.com).

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processor die using the face-to-face integration approach [4]. A silicon photonic chiplet is a die with prebuilt and preverified devices and circuits, which integrates optical devices. Instead of using monolithic integration, in our design, manufacturers can fabricate all the optical devices on CAMON. This can avoid many challenges in the integration of optical devices [5]. Besides, the reduction of die size can also increase the yield and lower the fabrication cost. Other contributions include the following.

1) A distributed control and arbitration system. In this system, the arbitration is made in a relatively independent way. The arbitration system can allocate the optical paths in a short time, even in large-scale systems.
2) A new structure of electrical–optical/optical–electrical (EO/OE) interfaces, which shows higher energy efficiency, lower conversion delay, and less area overhead than traditional EO/OE interfaces.
3) A preactivation mechanism for the off-chip laser source. If we put the laser to sleep at the idle time, this mechanism could help the system to activate the laser in advance. It makes the processor performance less sensitive to the laser activation latency.

The rest of this paper is organized as follows. Section II gives a survey. Section III introduces the state-of-the-art manycore processor. Section IV discusses CAMON. Section V quantitatively compares this paper with the traditional electrical network. Section VI concludes this paper.

II. Related Work

In the literature, many manycore processors use optical interconnects for on-chip communications. Vantrease *et al.* [6], Pan *et al.* [7], Cianchetti *et al.* [8], Shacham *et al.* [9], Joshi *et al.* [10], Beux *et al.* [11], Kao and Chao [12], Hamedani *et al.* [13], and Abellán *et al.* [14] proposed various types of optical NoCs for manycore processors. Li *et al.* [15], Bahirat and Pasricha [16], and Werner *et al.* [17] proposed hybrid NoCs, which combine both the optical network and the electrical network. In their systems, the optical network transmits part of the packets, and the electrical network transmits the rest part. Compared with the traditional electrical NoC, the optical NoCs or the hybrid NoCs show higher network throughput, lower latency, and lower power consumption. In addition to the on-chip interconnects, topologies also play an important role in the NoC. Networks with different topologies show different characteristics. Ramini *et al.* [18] compared different optical NoC topologies regarding the power consumption. Besides, Demir and Hardavellas [19] and Kurian *et al.* [20] proposed adaptive lasers. Vantrease *et al.* [6], Pan *et al.* [7], Li *et al.* [21], Zulfiqar *et al.* [22], and Werner *et al.* [23] proposed various arbitration systems for optical NoC.

Several manycore processors also use optical interconnects for interdie and interchip communications. Demir *et al.* [24] and Koka *et al.* [25] proposed optical networks connecting multiple dies within a package. The simulation shows that systems using optical interdie interconnects have better performance than systems using traditional interconnects. Studies from Hadke *et al.* [26], Morris *et al.* [27], Udipi *et al.* [28], Beamer *et al.* [29], and Batten *et al.* [30] show that by implementing an interchip optical network, the offered bandwidth between processor chip and memory modules can be effectively increased. The analyses show that optical interchip interconnect can be a promising candidate to alleviate the memory wall problem. Taking advantages of the optical interconnects, Abellán *et al.* [31], Xu *et al.* [32], Kurian *et al.* [33], and Grani and Bartolini [34] proposed various cache coherence protocols for optical networks, to further improve the performance of memory system. In some work, the optical NoC is integrated in the manycore processor using the 2.5-D technology. Bashir and Sarangi [35], Yin *et al.* [36], and Arunkumar *et al.* [37] proposed chiplet-based solutions for optical NoC. Grani *et al.* [38] and Werner *et al.* [39] proposed AWGR-based optical NoC using the 2.5-D technology.

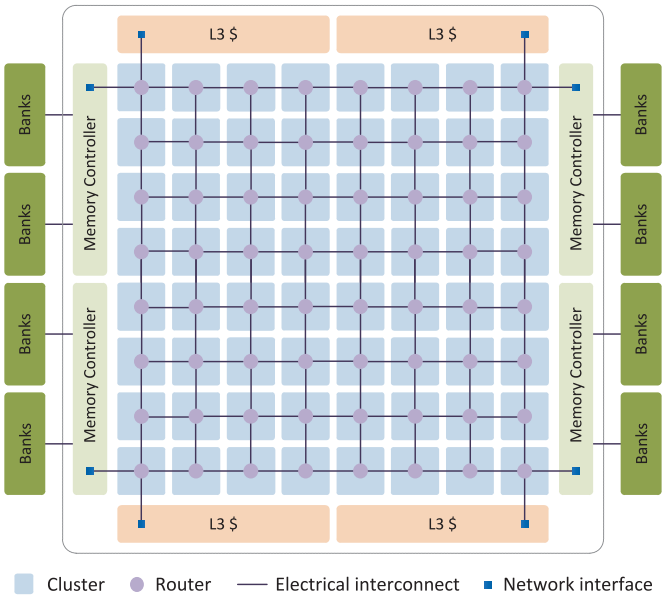


Fig. 1. Electrical network-based system with 64 clusters. Each cluster consists of two or four cores, private L1 caches, and a shared L2 cache.

III. State-of-the-Art Manycore Processor

Various manycore processors have been proposed and released, such as Intel's Xeon Phi (KNL) and Mellanox's TILE-Mx100. Fig. 1 shows a typical manycore processor. It group cores into clusters. A cluster contains two or four processor cores and one L2 cache. Each processor core has its private L1 data and instruction cache. The processor cores within the same cluster share the L2 cache in that cluster. All the processor cores share an L3 cache, which also serves as the last-level cache (LLC). The L3 cache slices and memory controllers are located at the edge of the processor chip. The clusters, directories, L3 cache slices, and memory controllers are attached to routers [40]. Electrical links connect all the routers into a mesh-based network. Many commercial products today use the mesh topology in their networks-on-chip. It is a good choice for the 128/256/512-core processors in terms of latency and area overhead. For example, in the 8×8 mesh network, the average hop count is 6.25, while the maximum radix of switches is only 5×5. Besides, the layout of mesh network is relatively simple with few crossover points. This feature could avoid large capacitance, which slows down the

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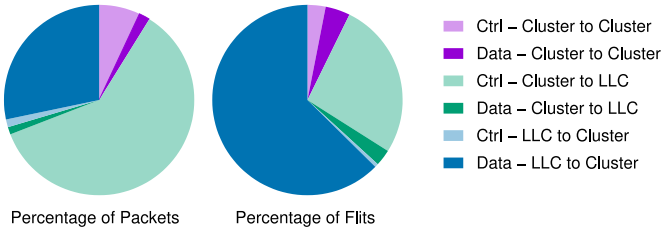


Fig. 2. Traffic distribution in the electrical network-based system regarding (a) packets and (b) flits. Statistics are from simulations on APEX benchmark [43].

circuit speed [41]. A packet consists of multiple flits. Every cycle, only one flit can be transmitted from a router to one of its neighboring routers. There are two types of packets in the network: 1) the data packet and 2) the control packet. The data packet is usually large, containing multiple flits, and the control packet is small, including only one flit. In some manycore processors, the memory controllers are connected to the off-chip memory modules by serial electrical interconnects [42], which can increase the memory bandwidth and improve the energy efficiency of data movement.

Manycore processors typically use the directory-based cache coherence protocol. The directory works as a lookup table, registering the coherence information of each L2 cache. During read operations, when a cache miss happens in one of the L2 caches, it will visit the directory first. If the L3 cache has the required data block, it will send a data packet to the L2 cache; if the valid copy of this data block is in another L2 cache, then the data packet is sent directly from that L2 cache. During write operations, if the L2 cache is going to modify a data block, it will read this data block first. Afterward, the directory will announce all the other L2 caches who also keep the copy of this data block, and set these data copies to be invalid. The directory has large area overhead [44], and is typically implemented on the processor chip. We find the traffic distribution among clusters and LLCs by simulating the manycore processor, shown in Fig. 2. This figure shows the average result of all the applications from the benchmark set APEX [43]. We assume that the directory is a part of the LLC. In some manycore processors, there is no L3 cache. In such case, the memory module works as the LLC. From the figure, on average, about 63% flits belong to data packets sent from the LLC to the cluster, and about 27% flits belong to control packets sent from the cluster to the LLC. An effective way to alleviate the communication bottlenecks is to accelerate the transmission of these two types of packets.

IV. Cache and Memory Optical Network

In this section, we show CAMON, a low-cost silicon photonic chiplet for manycore processors, which can effectively alleviate the communication bottlenecks of manycore processors.

A. Overview

We show the CAMON-based manycore processor in Fig. 3. Each cluster contains four cores/L1 caches and one L2 cache. Each processor core has its private L1 cache. The processor cores within the same cluster share the L2 cache in that

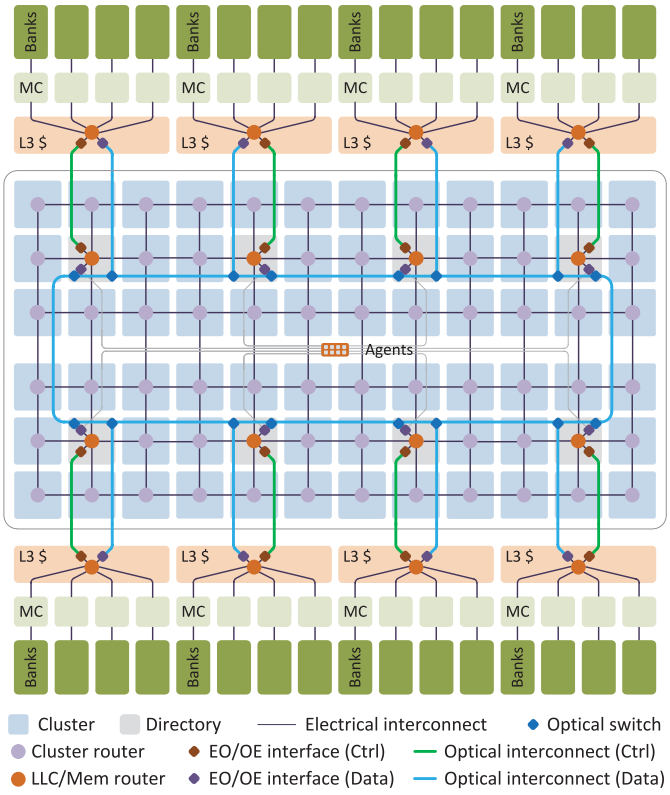


Fig. 3. Proposed CAMON-based system with 64 clusters. Each cluster consists of four cores, private L1 caches, and a shared L2 cache.

cluster. The directory is distributed over the processor die. Fig. 3 shows a 6×12 grid. Inside the grid, eight cells colored in gray represent the distributed directories, and the rest 64 cells colored in blue represent the clusters. We group the L3 cache slice, memory controllers, and memory banks into an LLC/Mem module. There are two types of routers: 1) cluster router and 2) LLC/Mem router. Clusters are attached to cluster routers, directories are attached to on-chip LLC/Mem routers, and LLC/Mem modules are attached to off-chip LLC/Mem routers. Electrical links connect the cluster routers and the on-chip LLC/Mem routers into a mesh-based network. Optical links connect the on-chip LLC/Mem routers and the off-chip LLC/Mem routers. In CAMON, the size of the L3 cache can be relatively large because it is not physically constrained by the processor package. When the memory access locality is high, increasing the size of the L3 cache can effectively reduce the cache miss rate, and thus reduce the average data fetching time for the processor core. On the other hand, when the memory access locality is low (when the application is reading a huge data set from the memory), the L3 cache has a very large miss rate whatever its size is. The data requested by the processor cores has a high chance to be only kept in the main memory. Whether the L3 cache is on the processor chip or not has limited effect to the memory access delay.

CAMON alleviate the traffic load by transmitting the data and control packets in two different optical communication systems. The first transmission system is a ring-based optical network connecting all the LLC/Mem routers (blue lines in Fig. 3). In this system, the optical channel is shared by multiple LLC/Mem routers. Before the transmission, each router needs

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to allocate the related optical channel from a controller, and the related switches should be turned on to set up the optical path. It is suitable for sending data packets. The second transmission system is a point-to-point connection between the on-chip and the off-chip LLC/Mem routers (green lines in Fig. 3). In this system, lights have only one direction to propagate, just from the source router to the destination router. Hence, there is no need to set up the optical path in advance. It is suitable for transmitting control packets. Each LLC/Mem router has two OE/EO interfaces connecting the two communication systems.

Data and control packets take different routing path (reasons discussed in Section IV-F). If the off-chip LLC/Mem module sends a data packet to a cluster, the routing algorithm first finds an on-chip LLC/Mem router that is closest to the destination cluster. Then the data packet is transmitted from the off-chip LLC/Mem router to that on-chip LLC/Mem router by the ring-based optical communication system. If the off-chip LLC/Mem module sends a control packet to a cluster, it is first transmitted from the off-chip LLC/Mem router to the on-chip LLC/Mem router by the point-to-point connection-based optical communication system. Then, the mesh-based electrical network transmits the packet to the destination cluster. Packet between two clusters, or packet between a cluster and a directory, no matter it is for data or control, is only transmitted by the mesh-based electrical network.

B. Photonic Chiplet

A silicon photonic chiplet is a die with prebuilt and preverified devices and circuits. It can integrate many types of optical devices, such as EO/OE interfaces, optical switches, waveguides, and optical pins. Using chiplets for manycore processor has two benefits: manufacturers can fabricate photonic devices separately; a smaller die size can increase the yield and lower the cost. By using photonic chiplet instead of monolithic integration, we can avoid many challenges in the implementation of optical devices [5]. In Fig. 4, we show two different photonic chiplets for the processor chip and the LLC/Mem module. They are mounted on top of the electrical dies using the face-to-face integration approach [4]. Optical pins are on the edge of the photonic chiplet. The state-of-the-art technology can support 125-μm pitch for optical pins [3]. The chiplet for the processor chip is bar-shaped, each of them integrates half of the optical network, and connects half of the on-chip LLC/Mem routers. Optical links connect two bar-shaped chiplets into a closed loop via optical pins. Another choice is to fabricate the entire optical network in a single chiplet. By rough estimation, the twin-chiplet design can save around 66% area, compared with the single-chiplet design. The chiplet for the LLC/Mem module is smaller, which connects to only one off-chip LLC/Mem router. Optical links connect the chiplet for the processor chip and the chiplet for the LLC/Mem module via optical pins.

C. Centralized Laser Bank

For easy maintenance and thermal control, CAMON uses a centralized off-chip laser bank as the light source. A laser bank consists of multiple lasers, which can emit lights in different wavelengths, as shown in Fig. 4. To save the power, CAMON

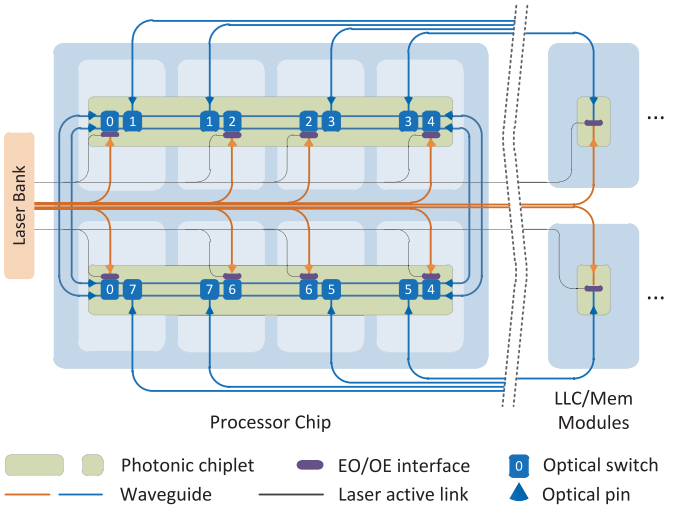


Fig. 4. Photonic chiplets for the processor chip and LLC/Mem modules. We put them on top of the electrical dies (blue squares). For simplicity reasons, we only show the ring-based optical network.

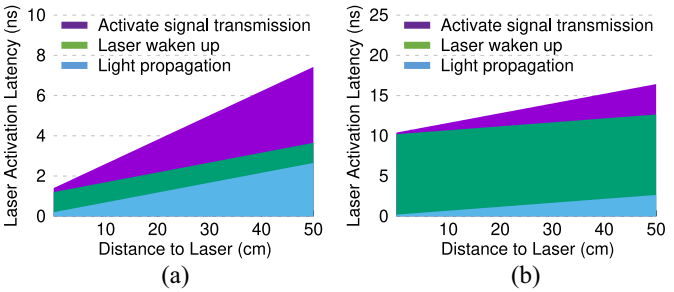


Fig. 5. Breakdown of the laser activation latency: (a) laser is waken up from standby mode and (b) laser is waken up from off mode.

puts lasers to sleep at the idle time. Each laser has three modes: 1) the full-on mode; 2) the standby mode; and 3) the off mode. If there is no packet to transmit, the laser is on the standby mode or the off mode. In the standby mode, lasers are driven by threshold currents. In the state-of-the-art technology [45], it takes about 1 ns to switch the laser from the standby mode to the full-on mode, and about 10 ns to switch the laser from the off mode to the full-on mode. Before transmitting packets, the control system will activate the related lasers to the full-on mode. It will increase the driving current of the laser. The exact current to drive the laser is precalculated based on the total power loss of the optical path. After transmission, the control system will switch the laser back to the standby mode or the off mode.

The total laser activation latency includes the time to wake up the laser, the transmission delay of the activation signal from the LLC/Mem router to the laser, and the propagation delay of light from the laser to the LLC/Mem router. We show the breakdown of the laser activation latency in Fig. 5. We can see that if the laser is switched from the standby mode, the total activation latency is typically less than 10 ns. If the laser is switched from the off mode, the total activation latency is typically less than 20 ns [2], [45]. In Section IV-I, we propose a laser activation mechanism to compensate this latency. Fig. 6 shows the details of the two optical communication

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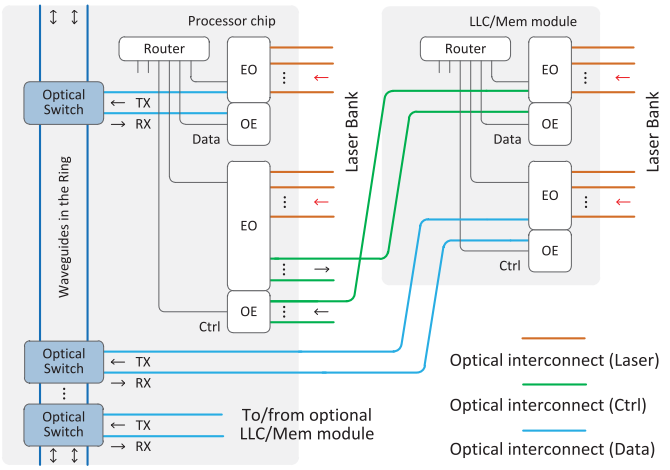


Fig. 6. Ring-based optical communication system (blue) for data packets and the point-to-point connection-based system (green) for control packets.

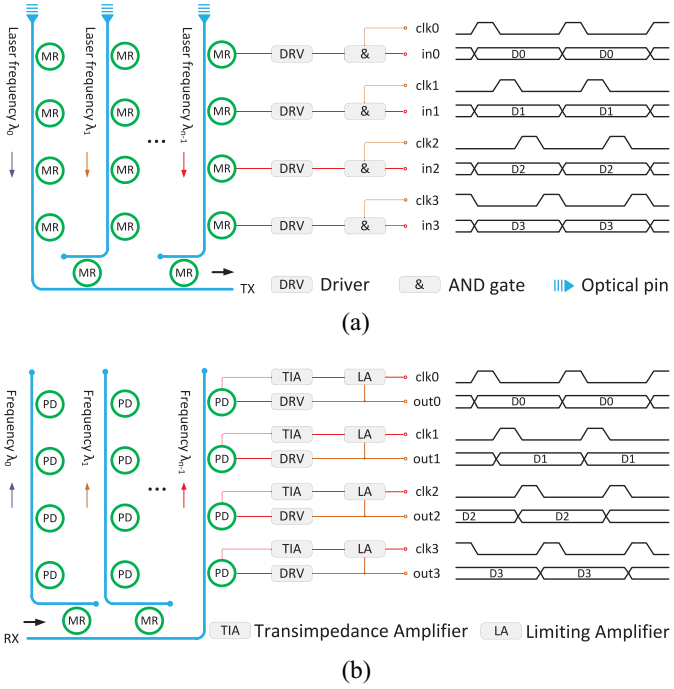


Fig. 7. Structure of (a) 4:1 EO interface and (b) 1:4 OE interface. The four clock signals are equally phase shifted by $\pi/2$.

systems. The red lines denote optical interconnects from laser sources, the blue lines denote the transmission system for the data packet, and the green lines denote the transmission system for the control packets. An EO interface modulates lights from off-chip lasers. After transmission, an OE interface converts optical signals back to electrical signals. A laser bank consists of multiple lasers, which can emit lights in different wavelengths. The laser bank is placed away from the many-core processor. Lights are generated from the laser bank and delivered to the manycore processor by fibers, and the processor chip sends activation/sleep signals to the laser bank via on-board electrical interconnect.

D. Low-Power Low-Latency EO/OE Interface

We propose an EO/OE interface for CAMON, which shows low conversion delay, high energy efficiency, and small area overhead. Traditional EO/OE interfaces use an electrical serializer/deserializer (SERDES) to do serialization/deserialization and a separated module to do EO/OE conversion. The whole process consumes a lot of time and energy. Our new interface can reduce the latency and save the energy because it can do serialization/deserialization and EO/OE conversion concurrently. We use microresonators (MRs) as modulation, switching, and filtering elements. In the EO interfaces, shown in Fig. 7(a), each wavelength of laser light is first modulated independently. Parallel electrical signals are converted to serialized optical signals by MRs. After modulation, multiple wavelengths of laser lights are switched into the transmit (TX) port by MRs. In the OE interface, shown in Fig. 7(b), each wavelength of laser light is first filtered out from the receive (RX) port by MRs. Serialized optical signals are converted to parallel electrical signals by MRs and photodetectors (PDs).

We design an optical SERDES to convert signals between parallel electrical formats and serial optical formats concurrently. In the E-O interface [Fig. 7(a)], there are four parallel electrical inputs, which are nonreturn-to-zero (NRZ) coded. The four clock signals are equally phase shifted by $\pi/2$. Each clock signal is used to control one of the electrical signals through a gate. The parallel signals are switched on by turns, and only one of the MRs is turned on at one time. Data from the four parallel electrical inputs are converted to a serial optical signal by those four MRs, in an interleaving pattern. In the O-E interface [Fig. 7(b)], each clock signal is used to turn on one of the MRs by turns. When the MR is turned on, it can filter the optical signals. The corresponding PD detects lights and generates the electrical signals, which is then amplified by the transimpedance amplifier and the limiting amplifier. In this way, data from the serial optical input are converted to four parallel NRZ electrical signals by four MRs. We use clock-forwarding technique to synchronize the receiver.

The traditional EO/OE interfaces contain multiple stages of latches, clock dividers, and multiplexers. The circuit in our proposed interface is simpler. We compare the two interfaces in terms of energy consumption per bit and area overhead. We use the method in [46] to model the power of electrical circuits in both interfaces. It is assumed that the supply current of a single gate running at full clock speed is I_0. For example, the supply currents of the multiplexer, the latch, and the clock divider are I_0, I_0, and $2I_0$, respectively. The supply currents of gates running at other clock speeds are scaled by their working frequencies. The areas of electrical circuits are compared by a similar method. We estimate the values of the unit current and the unit area based on statistics from fabricated prototypes [47]. It is assumed the data rate of each parallel signal is 2 Gb/s, and the 8 Gb/s, 16 Gb/s, and 32 Gb/s serial interconnects use the 4:1/1:4, 8:1/1:8, and 16:1/1:16 interfaces, respectively. From Fig. 8, we can see that if the data rate of the serial signal is 32 Gb/s, the proposed interface can save 77% energy consumption per bit and 79% area overhead on the electrical and silicon photonic dies, at 7-nm technology node. Besides, the optical interconnect with our proposed EO/OE interfaces is more energy efficient than the serial electrical interconnect. For example, with 32-Gb/s data rate, the energy

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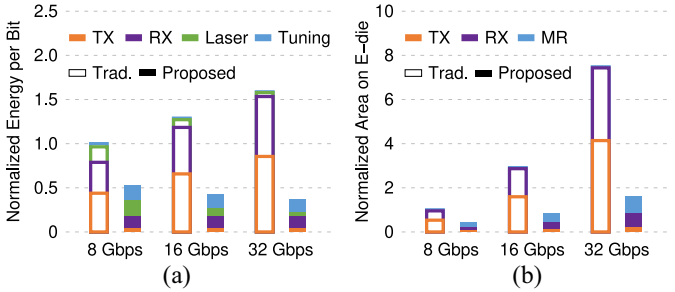


Fig. 8. (a) Normalized energy consumption per bit versus transmission data rate. (b) Normalized area on electrical die versus transmission data rate. Empty bar for traditional interface and filled bar for our proposed interface.

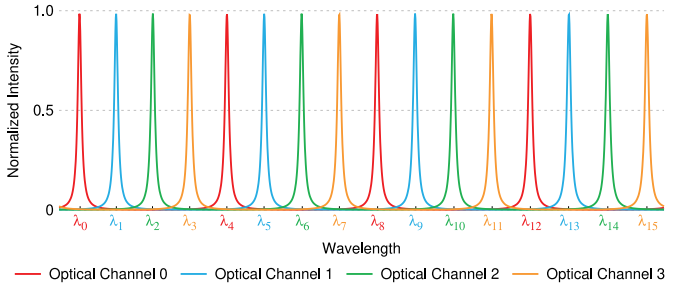


Fig. 9. Spectrum of four MRs. A single MR can switch lights in different wavelengths, which are grouped as an optical channel.

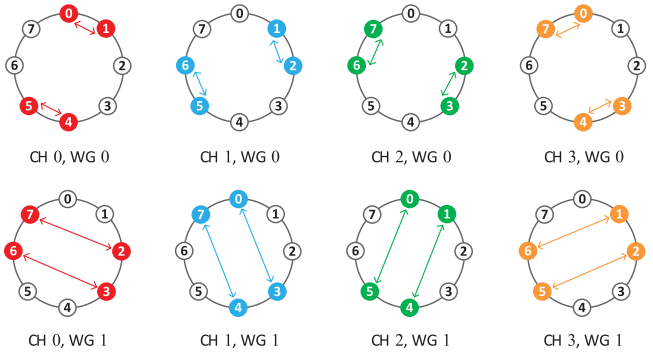


Fig. 10. Example mapping between the optical channel ID and the region ID. The nodes and arrows denote regions and channels, respectively.

consumption per bit of the optical interconnect is 76% less than that of the electrical interconnect.

E. Preassigned Optical Channel

We group multiple laser lights as one optical channel. The reason for the grouping is to reduce the number of MRs implemented in the network and thus decrease the total power loss. Fig. 9 shows the spectrums of four different MRs. We use a unique color to label each of them. From the spectrum, we can see multiple peaks, which are called resonance wavelengths. When we turn off the MR, its resonance wavelength is different from the wavelength of laser light. When we turn on the MR, its spectrum shifts and one of the resonance wavelengths may equal the wavelength of laser light. Under such circumstance, that laser light can be switched from one waveguide to another by the MR. If we well tune the wavelength of each laser light, one MR can switch multiple laser lights at the same time. We define an optical channel to be the group of laser lights that can be switched by the same MR. In Fig. 9, there are 16 laser lights in different wavelengths. They belong to four optical channels. Those four optical channels do not interfere with each other. They can be controlled by four MRs independently.

To effectively reduce the arbitration overhead and increase the network scalability, we preassign the optical wavelengths for the optical transmission. When an LLC/Mem router initiates an optical transmission, it only needs to allocate the corresponding optical wavelength. We divide the optical ring into multiple regions. The optical switches and LLC/Mem routers are located in different regions of the ring. Fig. 4 labels the region ID of each optical switch. There could be different mappings between the optical channel ID and the region ID of source and destination routers. A good mapping could well balance the cache coherence traffic loads. Based on the traffic patterns in Fig. 2, we show an example mapping in Fig. 10. There are two parallel waveguides in the optical ring. Each ring waveguide has four optical channels. Each node represents a region ID, and each colored arrow represents an optical channel. For example, if the LLC/Mem router at region 0 sends a packet to the LLC/Mem router at region 1, according to the mapping scheme shown in Fig. 10, the arbitration system will choose optical channel 0 in waveguide 0 for the data transmission.

In the CAMON-based system, we transmit packets between two clusters by the electrical mesh-based network and transmit packets between a cluster and an LLC/Mem router by the ring-based optical network. In this example, the on-chip LLC/Mem routers are located in a region with even ID, and the off-chip LLC/Mem routers are located in a region with odd ID. Hence, in Fig. 10, traffic only exists between a node with even ID and a node with odd ID. According to Fig. 2, read operations are more frequent than the write operations. There are two designs in the network that can balance the traffic load in the network. First, the address of the memory is interleaved. Therefore, the memory request has an equal chance to visit any of the memory modules. Second, by using our channel mapping scheme shown in Fig. 10, a single optical channel can handle about $1/N_c$ of the traffic from read operations and $1/N_c$ of the traffic from write operations, where N_c is the number of optical channels in the network.

We propose low-loss optical switches for CAMON, shown in Fig. 11. The total power loss incurred by MRs, including the passing loss and the insertion loss, is optimized [2]. Switches at different regions use different types of MRs, each type of MR can control only one optical channel, denoted by the color. For simplicity reasons, Fig. 11 only shows switches at region 0, 1, and 2. During the optical channel allocation, the agents send the configuration information to both the source and destination optical switches. After receiving the configuration information, the optical switch turns on related MRs, which enable the transmission between the ring waveguide (dark blue line) and the TX/RX port. Optical devices are sensitive to the environment temperature. To avoid thermal variations, we implement thermal tuning devices to control the temperature. We compare the optical network in CAMON with optical NoCs in other topologies, such as Butterfly, Benes, Fat-tree,

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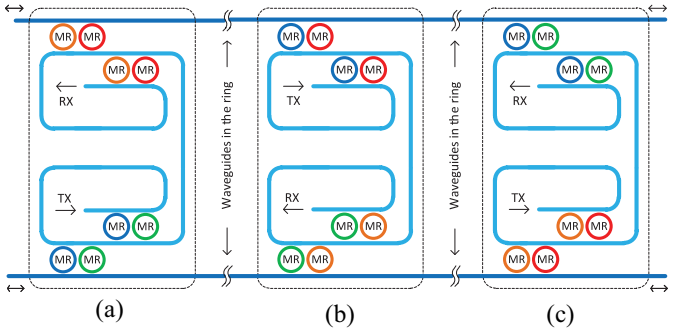


Fig. 11. Optical switches in different regions use different MRs. Each MR can switch only one optical channel, denoted by its color. (a) Switch at region 0. (b) Switch at region 1. (c) Switch at region 2.

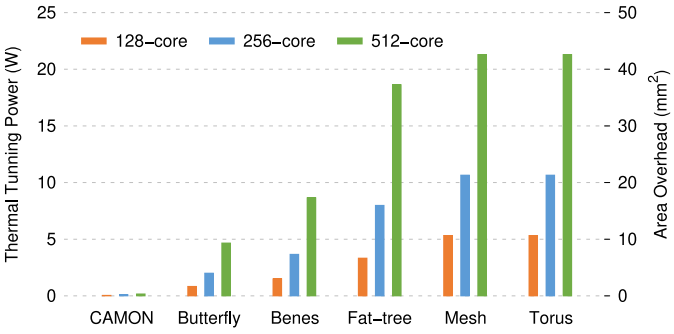


Fig. 12. Thermal tuning power and area overhead of all the basic optical switching elements (MRs), assuming four cores per cluster.

Mesh, and Torus. CAMON consumes the lowest thermal tuning power and has the least area overhead on photonic chiplets. From Fig. 12, we can see that on average, the thermal tuning power and area overhead of CAMON is only 0.8% of that in the mesh/torus-based optical NoC.

F. Diversion of Cache Coherence Traffic

To effectively reduce the network latency, we assign different routing paths for different cache coherence packets. We classify them into three categories based on Fig. 2. The first category includes data packets between the LLCs and the clusters. Packets in this category are usually large, and they place the most considerable burden on the network. To alleviate the situation, we transmit these packets through the ring-based optical communication system. The second category includes control packets between the LLCs and the clusters. When we transmit these packets, the interchip communication between the processor chip and the LLC/Mem module becomes a bottleneck. If we use electrical interconnects to transmit them from one chip to another, the number of pins on the package constrains the total bandwidth. To alleviate this bottleneck, we use the point-to-point connection-based optical communication system. The third category includes data and control packets between two clusters. Packets in this category place a relatively small burden on the network. We transmit these packets by the mesh-based electrical network.

G. Control and Arbitration System

We propose a distributed control and arbitration system for CAMON, which could avoid conflicts of channel allocation,

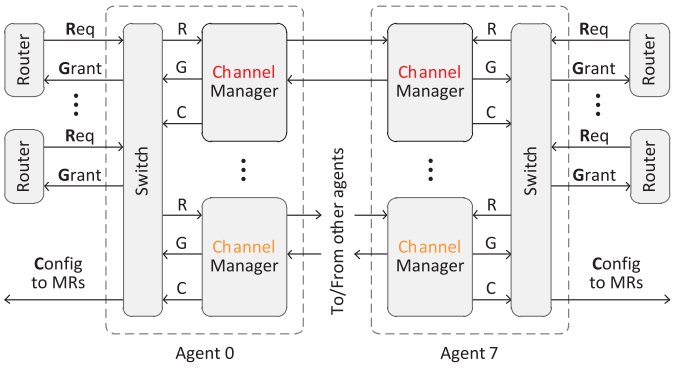


Fig. 13. Structures of two region agents. For simplicity reasons, we only show connections between the agent for region 0 and the agent for region 7.

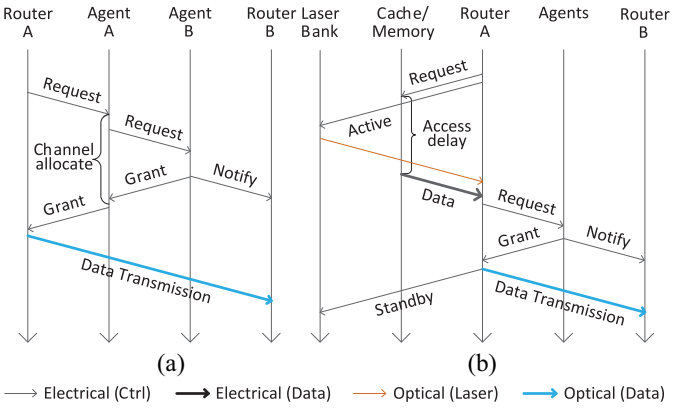


Fig. 14. Time-sequence diagrams of (a) optical channel allocation process and (b) laser preactivation mechanism.

and shorten the allocation time. If one LLC/Mem router sends a packet to another router using the ring-based optical network, it needs to allocate the optical channel first. Routers do not talk with each other directly. Instead, they communicate with an agent. We design a region agent for each region, which manages all the LLC/Mem routers in that region. The region agents are fully connected and located at the center of the processor chip. The on-chip LLC/Mem routers communicate with the agent by on-chip electrical interconnects, and the off-chip LLC/Mem routers communicate with the agent by optical interconnects.

Fig. 13 shows the structures of two region agents. For each channel that can be used by routers in this region, there is a channel manager inside the region agent. Based on the channel mapping (Fig. 10), the channel allocation request from the LLC/Mem router will be forwarded to the corresponding channel manager. The channel manager records the availability of the channel. If another router is using this channel, the channel becomes unavailable until that router finishes the transmission, and releases the channel. The manager also generates configuration information for MRs. Based on this information, the region agent can change the switching function of the optical switches. There is a timer inside each channel manager. After a certain amount of time, the channel is reset to be available. The time interval is precalculated based on the packet size and the transmission data rate.

Fig. 14(a) shows the arbitration algorithms. If the LLC/Mem router A sends a packet to router B via the ring-based optical

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network, it will first send a request signal to region agent A for channel allocation. The agent A then forwards this request to agent B. The two-region agents will check the availability of the optical channel. When the optical channel is available in both regions, a grant signal is sent back to router A, and a notify signal is sent to router B. The optical channel is allocated when the two LLC/Mem routers receive the notify or grant signals. At the same time, the two region agents change the switching function of the optical switches. If there are multiple path allocations requests, they are processed based on the first-come, first-served policy.

H. Deadlock Avoidance

We implement a deadlock avoidance mechanism. The deadlock happens if a router in region A sends a channel allocation request to a router in region B, and a router in region B sends an allocation request to a router in region A. Region A agent is waiting for the grant signal from region B agent, and at the same time, region B agent is waiting for the grant signal from region A agent. These two requests are in deadlock states. Under this circumstance, the system can never grant these two requests. To avoid any deadlocks in the ring-based optical network, we design two types of region agents: 1) master agent and 2) slave agent. We define the agents managing the off-chip LLC/Mem routers as master agents (odd ID), and the agents managing the on-chip LLC/Mem routers as slave agents (even ID). This can guarantee that the channel allocation request is always sent between a master agent to a slave agent.

The deadlock avoidance mechanism is shown in Fig. 15. Each request contains the ID of the source router and the ID of the destination router. If the master agent sends a request to the slave agent, the master agent will first check the source ID first and grant the request if the channel resource is available in the source region. Afterward, the slave agent will check the destination ID and grant the request if the channel resource is available in the destination region. On the other direction, if a slave agent sends a request to the master agent, the master agent will grant the request first and then followed by the slave agent. No matter in which direction, the master agent always grants the request first. Multiple requests in the same channel manager are processed by the master agent successively based on the first-come, first-served policy.

I. Laser Preactivation Mechanism

We propose a laser preactivation mechanism for CAMON, which could compensate for the delay of laser activation, and reduce the initialing time for optical transmission. This mechanism can be used in both read and write operations. Because for either operation, we will read the data first from a cache or memory module. After transmission, the destination cache or memory module receives the data and writes them into their banks. Fig. 14(b) shows this mechanism. When an LLC/Mem router receives a read request first, it sends the request to the related cache or memory module, and at the same time sends an activation signal to related lasers. It takes time for the cache or the memory to fetch data from its bank. It also takes time to activate the laser because they are off-chip. Normally, the laser activation latency of laser is less than 10 ns [2], [45], and

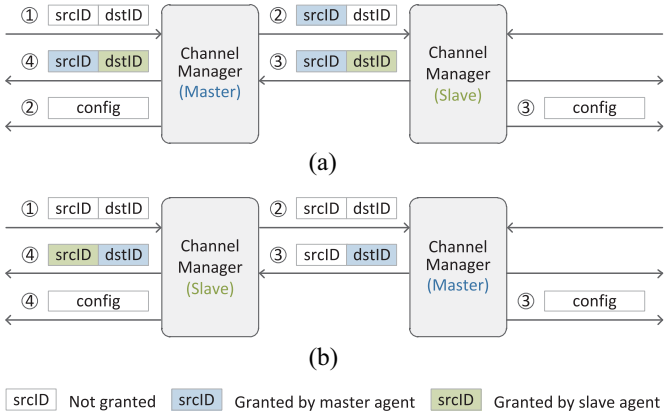


Fig. 15. Deadlock avoidance mechanism. A request contains the ID of the source LLC/Mem router and the ID of the destination router. (a) Allocation from a master agent. (b) Allocation from a slave agent.

the data comes to the router later than the light. In such case, the router immediately sends a channel allocation request to the region agent. If the data is already in the I/O buffer of the bank, the data may come to the router early than the activation of the laser source. Under this circumstance, the router does not send the channel allocation request until the light comes to the router. Because it takes longer to read data from the LLC/Mem module than the L2 cache, the on-chip LLC/Mem router would be more sensitive to the laser activation latency than the off-chip LLC/Mem router. Therefore, we put the laser bank closer to the processor chip (Fig. 4) in order to reduce the activation latency of the laser for the L2 cache.

V. Quantitative Analysis and Comparison

In this section, we compare the CAMON-based system and the electrical network-based system regarding performance, energy consumption, and area overhead.

A. Experimental Setup

We evaluate the performance of the CAMON-based system and the electrical network-based system by a full-system simulator JADE [48]. It uses the Ruby memory system [49] to model the cache coherence and memory behavior. Table I shows the configuration of the manycore processor. McPAT [50] generates the power consumption and area of processor cores, caches, directories, and memory controllers. We obtain the energy consumption and area of region agents by synthesizing them using Synopsys tools. We calculate the power consumption of memory modules based on Micron's memory model [51], and the energy consumption of electrical and optical interconnects by OEIL [2]. The power consumption of optical interconnects includes the power of laser sources, modulators, receivers, and thermal tuning devices. We calculate the power consumption of laser sources by finding the worst-case power loss of any possible optical channels. We implement heaters on the silicon photonic chiplet to keep the working temperature of optical devices stable. Each MR is equipped with a photocurrent monitor, an EO tuner, and a micro heaters [52]. These thermal tuning devices are working at all times. Even if the MR is not used in switching, modulating, or filtering lights, they keep working and consuming energy. The parameters of optical devices are collected

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TABLE I
SYSTEM CONFIGURATION

Parameter	Value
Processor core	ARMv8 core, @2∼4GHz, 4 cores/cluster
L1 I/D cache	64 KB/core, private
L2 cache	2 MB/cluster, shared by cores in one cluster
L3 cache	64 MB/module, shared by all cores
Directory	Distributed, keep L2 coherent
Cache coherence	Directory-based MESI, 64 Byte/block
Memory	8 GB/module, FR-FCFS schedule policy
Router parameter	16 Byte flit size, 4-stage pipeline
Technology	7 nm logic die, 12 nm memory die

TABLE II
OPTICAL DEVICE PARAMETER

Parameter	Value
MR passing (input port to through port)	0.05 dB
MR insertion (input port to drop port)	2 dB
MR heat tuning power	0.65 mW
Waveguide propagation loss	2 dB/cm
Fiber propagation loss	5×10^{-6} dB/cm
Optical pin coupling loss	1.5 dB
Receiver sensitivity	-20 dbm $\sim$ -12 dB
Laser power conversion efficiency	15% $\sim$ 30%
Data rate per wavelength	16∼32 Gbps

from [53]–[55], and summarized in Table II. We show a range of values for some parameters.

We evaluate the system performance using the application performance at extreme scale (APEX) benchmark [43], and the COSMIC heterogeneous multiprocessor benchmark [48]. The APEX is an architecture independent performance characterization and benchmark set for scientific applications. Both APEX and COSMIC are developed from the algorithms of typical manycore processor applications. These two benchmark sets cover both memory intensive and nonmemory intensive applications, including many widely used applications, such as machine learning. Fig. 2 shows that the L1 and L2 caches on the chip could not hold the entire working set of the applications. The electrical network-based system has a similar system configuration as the CAMON-based system. Fig. 1 shows a 256-core example. The topology of the electrical network is mesh. For a fair comparison, we assume that the electrical network-based system uses serial electrical interconnects in the memory port [42], which provides much higher memory bandwidth than traditional memory systems. The serial interconnect-based system supports large memory capacities by connecting multiple memory modules into a memory network, which share the same memory port of the processor chip. We evaluate three different system scales: 1) 128-core; 2) 256-core; and 3) 512-core. It is assumed that the number of cores per cluster is fixed at any scale. Hence, the number of clusters is proportional to the number of processor cores. These systems are assumed to work at three different operation frequencies: 1) 2 GHz; 2) 3 GHz; and 3) 4 GHz. The data rates of the on-chip and the off-chip interconnects are scaled based on the operating frequency.

B. Performance and Performance Per Energy

Fig. 16 shows the normalized performance of the CAMON-based system and the electrical network-based system. From the figure, the CAMON-based system shows better performance than the electrical network-based system, particularly for large-scale systems. In average, the CAMON-based system shows 1.9× speedup for all the applications. Among the three system scales, the 512-core system shows the most considerable 2.7× speedup. Compared with the traditional electrical network, CAMON is more suitable for large-scale systems because of its capability of handling heavy traffic loads. For memory intensive applications (excluding ML-FMP), on average, the CAMON-based system shows 2.3× speedup. The CAMON-based system not only shows better performance but also higher energy efficiency than the electrical network-based system. Fig. 17 shows the normalized performance per energy of these two systems. Performance per energy measures the rate of computation that can be delivered by the processor system for every unit of energy consumed. This value is inversely proportional to the energy-delay product (EDP). It should be noted that in this experiment, we evaluate the energy consumption of the entire system, including the processor cores, caches, memory modules, and the network. From the figure, compared with the electrical network-based system, the CAMON-based system shows 2.9× improvement on performance per energy. Among the three system scales, the 512-core system shows the most considerable 4.6× improvement. For memory intensive applications, the performance per energy of the CAMON-based system is 4.2× the value of the electrical network-based system. All these improvements indicate the advantages of using CAMON for manycore processors.

The number of the laser is one factor that may affect the performance of the CAMON-based system. To reduce the static power consumption, we try to minimize the number of lasers in the CAMON-based system and increase the utilization of each laser. However, if we use less number of lasers, the data transmission bandwidth is decreased, and thus it takes more time to transmit a packet in the optical communication systems. Fig. 18(a) shows the normalized performance of the CAMON-based system versus the number of lasers. From the figure, we can see that if we increase the number of lasers from 1 laser/cluster to 2 laser/cluster, the performance of the CAMON-based system can be improved by 37%. However, if we further increase the number of lasers from 2 laser/cluster to 8 laser/cluster, the performance of the CAMON-based system can be improved by only 1%. We think that 2 laser/cluster could be an optimized point for the tested applications, and we assume this value in the other experiments.

The activation latency of lasers is another factor that may affect the performance of the CAMON-based system. CAMON uses an off-chip laser bank. Increasing the distance between the laser bank and the processor chip or the LLC/Mem module will increase the activation latencies of lasers, and thus increase the initiating time for optical transmission. Fig. 18(b) shows the normalized performance of the CAMON-based system versus the laser activation latency. From this figure, we can see that when the laser activation latency increases from 2.5 to 10 ns, it is hard to observe any

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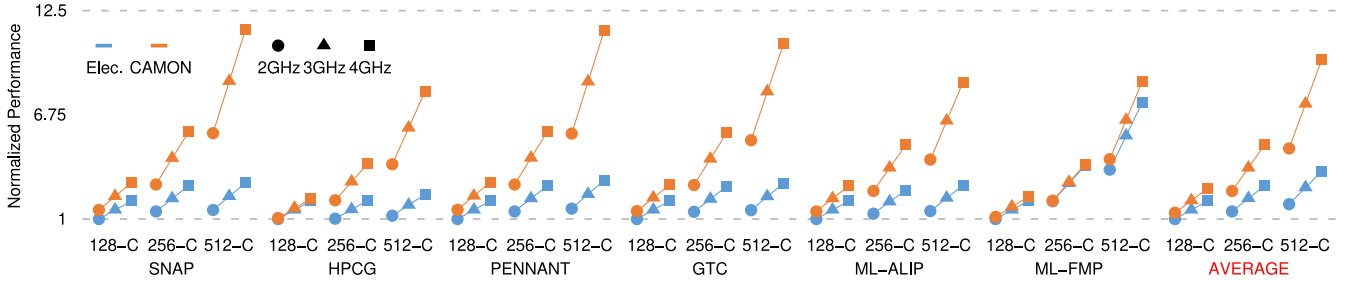


Fig. 16. Normalized performance of the CAMON-based system and the electrical network-based system versus system scale and operation frequency. For each application, we normalize the results by the performance of the 128-core and 2-GHz electrical network-based system.

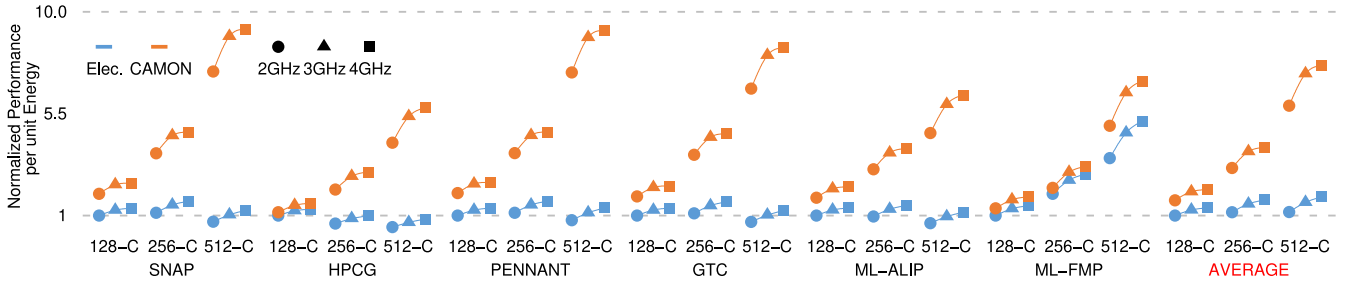


Fig. 17. Normalized performance per energy of the CAMON-based system and the electrical network-based system versus system scale and operation frequency. For each application, we normalize the results by the performance per energy of the 128-core and 2-GHz electrical network-based system.

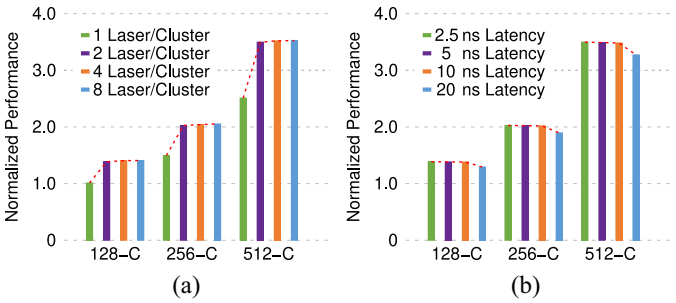


Fig. 18. Normalized performance of the CAMON-based system versus (a) number of lasers and (b) laser activation latency, in average. We normalize the results by that of the electrical network-based system.

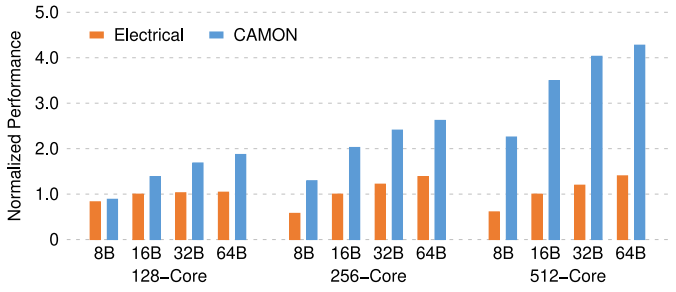


Fig. 19. Normalized performance of the CAMON-based system and the electrical network-based system versus flit size, in average. We normalize the results by that of the electrical network-based system in 16-Byte flit size.

performance degradation (<0.4%). Even if the latency further increases from 10 to 20 ns, the performance of the CAMON-based system only shows 6% degradation. Typically, we can keep the laser activation latency within 10 ns in the board level. Therefore, we can conclude that by using the preactivation laser mechanism, the performance of the CAMON-based system is insensitive to the location of the laser source.

C. Bandwidth and Latency

The bandwidth of the electrical network is related to the flit size. On the one hand, higher flit size means higher network bandwidth. On the other hand, higher flit size brings about higher energy consumption and area overhead of the routers. The exact size of the flits size is the tradeoff between the network performance and the cost [56]. In the state-of-the-art systems, the typical flit size is 16 Byte. We compare the performance of the CAMON-based system and the electrical network-based system versus flit size. From Fig. 19, we can see that, in average, compared with the electrical network-based system, the CAMON-based system shows 2.30× speedup in 16-Byte flit size and 2.24× speedup in 64-Byte flit size. We can observe similar speedup from the CAMON-based system, with increased bandwidth in the electrical network.

The high bandwidth of CAMON is one factor that can speedup the manycore processor. Fig. 20 shows the average memory bandwidth of the CAMON-based system and the electrical network-based system. The memory bandwidth is the rate at which data can be read from or write into the memory banks. Based on the delivered memory bandwidth, we classify the applications into two categories. In this experiment, most applications show large delivered memory bandwidth. They are memory intensive applications. One application (ML-FMP) shows small delivered memory bandwidth. It is a nonmemory intensive application. From the figure, the CAMON-based system shows larger bandwidth than the electrical network-based system at any scale or any operating frequency. In this experiment, the largest memory bandwidth that can be delivered by the CAMON-based system is around 0.52 Tbps. For memory intensive applications, the average memory bandwidth of the CAMON-based system is 2.3× that of the electrical network-based system. Because

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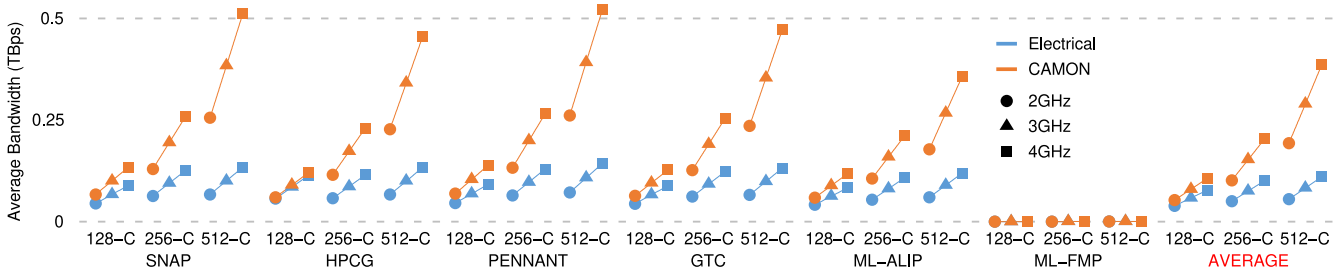


Fig. 20. Average memory bandwidth of the CAMON-based system and the electrical network-based system versus core count and operation frequency.

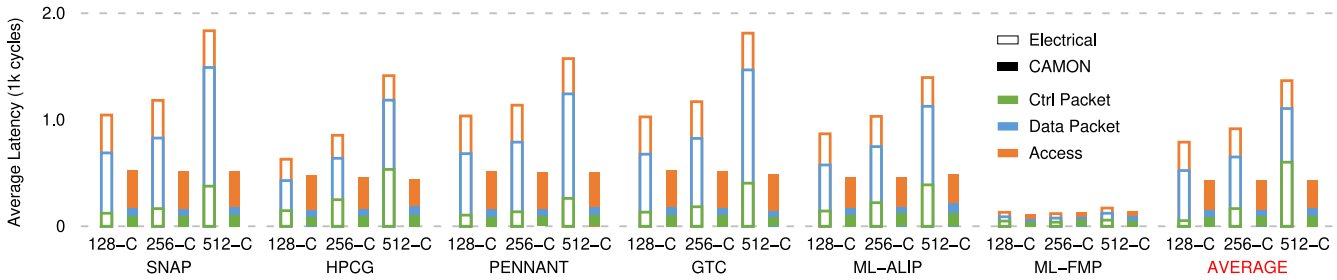


Fig. 21. Average latency in the CAMON-based system (filled bar) and the electrical network-based system (empty bar) versus core count.

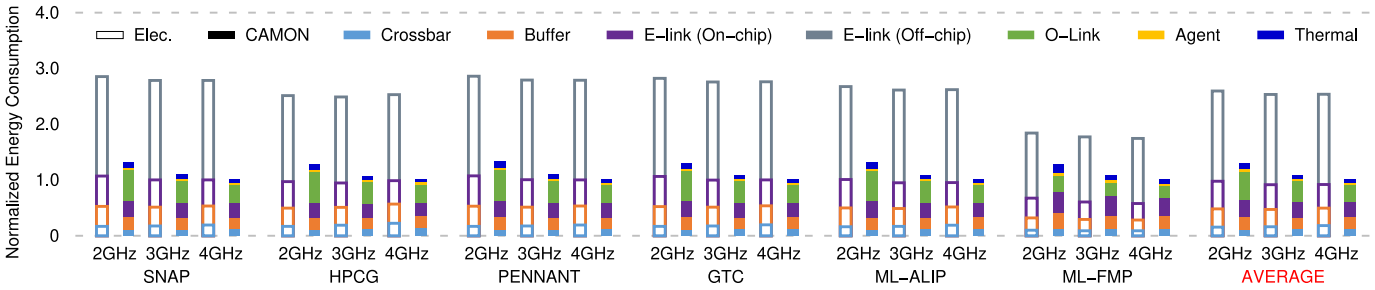


Fig. 22. Energy consumption breakdown of the CAMON (filled bar) and the electrical network (empty bar) in a 256-core system. For each application, we normalize the results by the energy consumption of the CAMON working at 4 GHz.

of the high bandwidth, CAMON is able to transmit a large volume of data packets more efficiently than the electrical network. The waiting time for each package in the router is reduced. As a result, CAMON could effectively alleviate the traffic bottlenecks, and speedup the manycore processor.

The low latency of CAMON is another factor that can speedup the manycore processor. Fig. 21 shows the average read latencies in the CAMON-based system and the electrical network-based system, after the L2 read miss. The total latency includes three parts: 1) the control packet delay; 2) the data packet delay; and 3) the LLC/Mem access delay. The control or data packet delay is the total number of cycles spent by the network to transmit control or data packets. The access delay is the total number of cycles spent by the LLC or the memory module to fetch date block from their banks. From the figure, the CAMON-based system shows less latency than the electrical network-based system at any scale. On average, the total latency in the CAMON-based system is about 48% less than that in the electrical network-based system. For the data packet, the average latency in the CAMON-based system is about 77% less than that in the electrical network-based system. In the CAMON-based system, the ring-based optical network handles most long-distance transmission of data packets. Although it takes some time to set up the path, the optical network has large bandwidth, and it can send the data packet from the source router to the destination router directly, making the overall latency of the CAMON less than the electrical network. The network in CAMON is based on the ring topology. Given the fact the propagation speed of optical signals is about 20 cm/ns, the traversal time in the optical ring has close values in terms of cycles, whatever the addresses of the source and the destination routers are. Assuming the latency for each individual request is L_r, and each processor core is able to issue N_r requests at the same time, from the system perspective, the effect latency of each memory access is only L_r/N_r. Given the fact that the maximum value of N_r is 16 in the state-of-the-art manycore processor, the effective latency L_r/N_r is much smaller than L_r.

D. Energy Consumption

CAMON can improve the energy efficiency of manycore processor mainly by two reasons. First, CAMON itself consumes less energy than the electrical network. Fig. 22 shows the energy consumption breakdown of the CAMON and the electrical network. The energy consumption of the optical links includes the static power of laser source on standby

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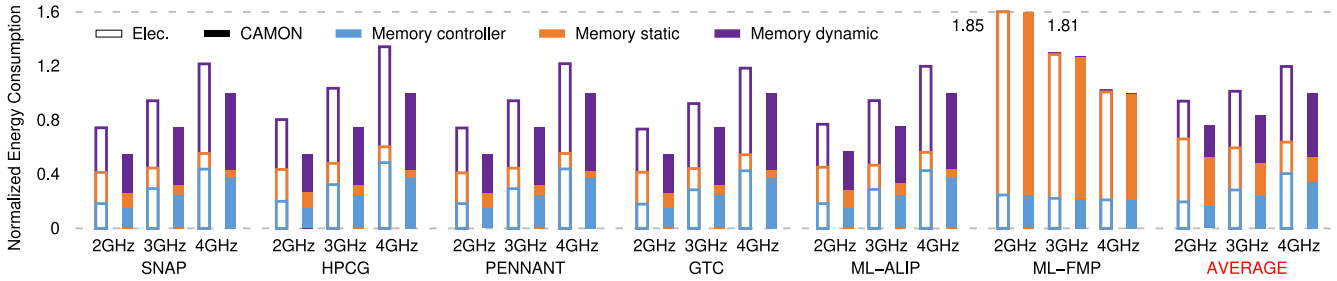


Fig. 23. Energy consumption breakdown of the memory modules in the 256-core CAMON-based system (filled bar) and the electrical network-based system (empty bar). For each application, we normalize the results by the energy consumption of the memory modules in the CAMON-based system at 4 GHz.

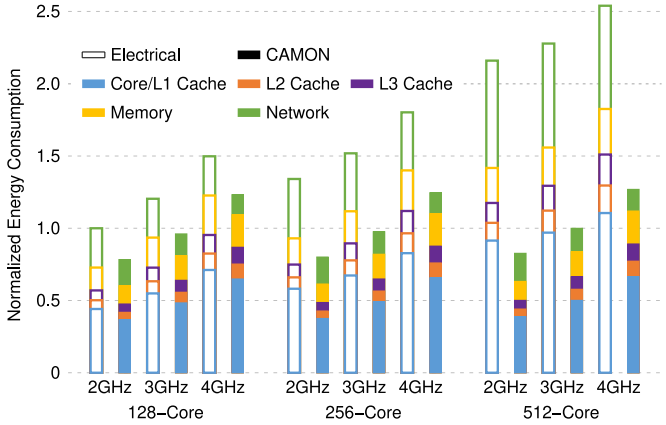


Fig. 24. Normalized energy consumption breakdown of the CAMON-based system (filled bar) and the electrical network-based system (empty bar).

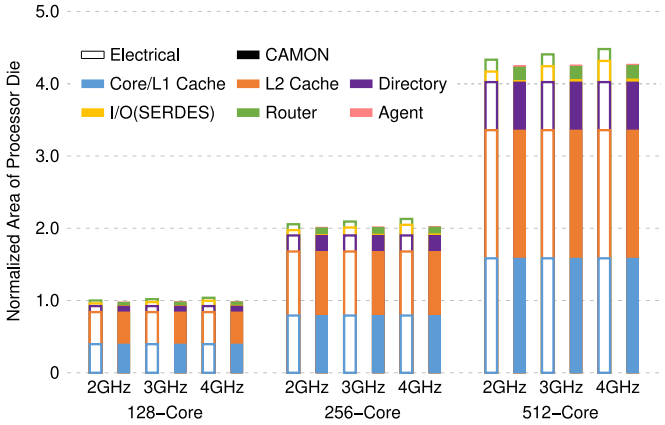


Fig. 25. Normalized area breakdown of processor dies in the CAMON-based system (filled bar) and the electrical network-based system (empty bar).

mode. From the figure, we can see that on average, the energy consumption of the CAMON is 55% less than that of the electrical network. Notably, the energy consumption of optical interconnects (including energy consumption on thermal tuning devices and agents) is 68% less than that of the off-chip electrical interconnects. The majority of energy saving comes from the SERDES circuit. It is because the EO/OE interfaces in the CAMON-based system use optical devices to do serialization/deserialization. Compared with the traditional SERDES used in the electrical network-based system, this new interface can save much energy. Second, the processors, caches, and memory modules consume less energy on static power because CAMON could reduce the running time of the whole system. For example, Fig. 23 shows the energy consumption breakdown of the memory modules in the CAMON-based system and the electrical network-based system. On an average, the energy consumption of the memory modules in the CAMON-based system is 19% less than that in the electrical network-based system. We can see that the majority of energy saving comes from the static energy of memory banks (42% energy reduction).

Fig. 24 shows the energy consumption of the CAMON-based system and the electrical network-based system. In average, the CAMON-based system shows 37% less energy consumption than the electrical network-based system. Figs. 17 and 24 together give us suggestions on the choice of the operation frequency and the scale of the CAMON-based system. On the one hand, systems with lower operation frequency or smaller core count show lower energy consumption. On the other hand, systems with higher operation frequency or larger core count show higher performance per energy. For example, the 2-GHz and 128-core CAMON-based system shows 62% higher energy efficiency, but 78% lower performance per energy than the 4-GHz and 256-core CAMON-based system. In theory, the energy consumption equals tP, where t is the execution time and P is the full-system power consumption. The performance per energy is proportional to $1/tE$, where E is the full-system energy consumption. From the above equations and figures, we can conclude that for the CAMON-based system: if the power budget is the primary consideration, lower operation frequency or smaller core count is preferred; if the time budget is the primary consideration, higher operation frequency or larger core count is preferred.

E. Area

Fig. 25 shows the area of the processor die in the CAMON-based system and the electrical network-based system. In the CAMON-based system, we implement the LLC outside the processor chip. Some state-of-the-art manycore processors place the LLC on the processors dies while others not. They either embed the LLC in the processor package or put it on board close to the processor chip. For a fair comparison, we do not include the area overhead of the LLC in the total area of the processor die. From the figure, although the LLC/Mem routers and the region agents have additional area overhead, CAMON can still reduce the area of the processor die by 3%. The majority of area saving comes from the

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SERDES circuit. In average, the area of the SERDES circuit in the CAMON-based system is about 85% less than that in the electrical network-based system. It is because the EO/OE interfaces in the CAMON-based system use optical devices to do serialization/deserialization. Compared with the traditional SERDES used in the electrical network-based system, this new interface can save much space on the die.

VI. Conclusion

In this paper, we proposed CAMON, a silicon photonic chiplet for manycore processors. It can effectively alleviate the communication bottlenecks of manycore processors and improve the energy efficiency of data movement, especially for large-scale systems. We propose several techniques for CAMON, including a distributed control and arbitration system, a low-power low-latency EO/OE interface, and an off-chip laser preactivation mechanism. The experimental results show that compared with the traditional electrical network, CAMON can improve the full-system performance per energy by 4.6×, speedup the manycore processor by 2.7×, and reduce the area of the processor die by 3%, in a 512-core system.

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Zhehui Wang (S'12–M'17) received the B.S. degree in electrical engineering from Fudan University, Shanghai, China, in 2010, and the Ph.D. degree in electronic and computer engineering from the Hong Kong University of Science and Technology, Hong Kong, in 2017.

Since 2017, he has been a Visiting Scholar with the Hong Kong University of Science and Technology. His current research interests include multiprocessor system, memory system, network-on-chip, optical interconnect, SERDES design, and floorplan design.

Zhifei Wang received the B.S. degree from Zhejiang University, Hangzhou, China, in 2014. He is currently pursuing the Ph.D. degree with the Hong Kong University of Science and Technology, Hong Kong.



Jiang Xu (S'02–M'07) received the Ph.D. degree from Princeton University, Princeton, NJ, USA, in 2007.

From 2001 to 2002, he was a Research Associate with Bell Labs, Murray Hill, NJ, USA, and with NEC Laboratories America, Princeton, NJ, USA, from 2003 to 2005. He joined a startup company, Sandbridge Technologies, New York, NY, USA, from 2005 to 2007 and developed and implemented two generations of NoC-based ultralow power multiprocessor systems-on-chip for mobile platforms. He is a Full Professor with the Hong Kong University of Science and Technology, Hong Kong, where he is the Founding Director of Xilinx-HKUST Joint Lab and established Big Data System Lab and Optical/Photonic Technology for Interconnected Computing System Lab. He has authored over 120 book chapters and papers in peer-reviewed journals and international conferences. His current research interests include big data system, heterogeneous computing, optical interconnection network, power delivery and management, MPSoC, low-power embedded system, and hardware/software codesign.

Dr. Xu currently serves as an Associate Editor for the IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS and the IEEE TRANSACTION ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS, and an Area Editor of Hardware and Architecture Design for the *ACM Transactions on Embedded Computing Systems*. He was an IEEE Distinguished Lecturer and an ACM Distinguished Speaker. He served on the steering committees, organizing committees, and technical program committees of many international conferences.

Yi-Shing Chang (M'01) received the Ph.D. degree from the Department of Electrical Engineering Systems, University of Southern California, Los Angeles, CA, USA, in 2001.

He joined Design and Test Technology, Intel, Folsom, CA, USA, upon graduation in 2001. He was involved in developing various deep submicron noise testing solutions, including crosstalk and power supply noise, and defect-based testing methodologies, including delay, transition, and logic proximity bridge faults for microprocessors and SoC. From 2005 to 2009, he researched on test selection, scan-out selection, CPU diagnosis, and test program offline validation. A full software suite was developed for test program offline validation and enabled virtual device under test offline simulation for SoC and CPU to catch test program bugs without running thousands of devices on the testers. From 2010 to 2013, he led the developing of tester runtime library for in-house testers and it gained project of record for all Intel SoC projects. In 2014, he transferred to Silicon Photonics Product Division, DCG, Santa Clara, CA USA, responsible for production test solutions, where he is a Principal Engineer responsible for all software development, test automation, and database solutions. He has authored or coauthored over 20 papers in peer reviewed internal and international conferences. His current research interests include big data system for high volume production, machine learning for manufacturing, and optical transceiver and interconnect system.

Dr. Chang was an SRC Mentor and served as a Technical Program Committee Member for ATS from 2013 to 2015 and ICCD from 2014 to 2017.

Jun Feng received the B.S. degree from Nanjing University, Nanjing, China, in 2017. He is currently pursuing the Ph.D. degree with the Hong Kong University of Science and Technology, Hong Kong.

Xuanqi Chen received the B.S. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 2015. He is currently pursuing the Ph.D. degree with the Hong Kong University of Science and Technology, Hong Kong.

Shixi Chen received the B.S. degree from the Hong Kong University of Science and Technology, Hong Kong, in 2018, where he is currently pursuing the Ph.D. degree.

Jiaxu Zhang received the B.S. degree from Zhejiang University, Hangzhou, China, in 2018. He is currently pursuing the Ph.D. degree with the Hong Kong University of Science and Technology, Hong Kong.

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